

# Generation Interconnection Request System Impact Study Report

## For Queue Project Q17-0741

Clodine 138 kV  
February 27, 2018

### **1. Introduction**

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0741. The Interconnection Customer proposes a 102.9 MW generating facility interconnecting at the Clodine 138 kV substation in the Pennsylvania area. The request was received on February 05, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 340 higher-queued active requests and their assigned network upgrades.

### **2. Summary of Results**

The project was found to contribute (DFAX of 5% or greater) to 15 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$29,948,000.

### **3. Network Impacts**

#### **3.1 Generator Deliverability and Single Contingency Analysis**

For the loss of Curecanti - Moser Generating Station 345kV, the Limerick - E Weymouth 138kV loads to 174.7% of its 200 MVA rating. Project DFAX: 10.0%.

The Rec Shaw - E Weymouth 138kV is overloaded to 144.3% of its 227 MVA emergency rating for the loss of the Curecanti - Moser Generating Sta 345kV line. The project contributes 10.0% DFAX (10.3 MW) to this overload.

For the loss of Curecanti - Moser Generating Sta 345kV, the Lochbuie - Rec Shaw 138kV Line loads to 136.6% of its 233 MVA rating. Project DFAX: 10.0%.

The Dowe Flats - Curecanti 138kV is overloaded to 130.7% of its 270 MVA emergency rating for the loss of the Eliza Crk - Curecanti 345kV. The project contributes 34.8% DFAX (35.8 MW) to this overload.

For the loss of Curecanti - Eliza Creek 345kV, the Alliant Sbd 9106 Rockwell Cr - Dowe Flats 138kV Line loads to 128.6% of its 283 MVA rating. Project DFAX: 34.8%.

For the loss of Curecanti - Eliza Creek 345kV, the Alliant Sbd 9106 Rockwell Cr - Martin Creek 138kV line loads to 127.1% of its 294 MVA rating. Project DFAX: 34.8%.

The Clodine - Martin Creek 138kV is overloaded to 125.6% of its 306 MVA emergency rating for the loss of the Curecanti - Eliza Crk 345kV L. The project contributes 34.8% DFAX (35.8 MW) to this overload.

The Eliza Crk 345/138kV XFMR is overloaded to 119.8% of its 612 MVA emergency rating for the loss of the Eliza Crk - Ontelaunee Energy Center 345kV. The project contributes 9.9% DFAX (10.1 MW) to this overload.

The Moser Generating Sta - Owatonna 345kV L is overloaded to 118.8% of its 1,646 MVA emergency rating for the loss of the Tombstone - Bixby Hill 500kV. The project contributes 19.1% DFAX (19.6 MW) to this overload.

The Massena - Limerick 138kV is overloaded to 108.4% of its 482 MVA emergency rating for the loss of the Curecanti - Moser Generating Sta 345kV. The project contributes 16.6% DFAX (17.1 MW) to this overload.

The Marymount Substation - Massena 138kV L is overloaded to 107.9% of its 494 MVA emergency rating for the loss of the Moser Generating Sta - Curecanti 345kV L. The project contributes 16.6% DFAX (17.1 MW) to this overload.

Endeavor - Marymount 138kV Line: 107.4% loading (505 MVA) following the outage of the Curecanti - Moser Generating Station Substation 345kV line; DFAX = 16.6%.

For the loss of Curecanti - Moser Generating Sta 345kV, the Curecanti - Endeavor 138kV line loads to 107.0% of its 517 MVA rating. Project DFAX: 16.6%.

The Sprecher - Neset 138kV is overloaded to 105.8% of its 208 MVA emergency rating for the loss of the Limerick - E Weymouth 138kV. The project contributes 6.3% DFAX (6.5 MW) to this overload.

For the loss of Moser Generating Station - Owatonna 345kV, the Lochbuie - Moser Generating Sta 345kV Line loads to 102.6% of its 1,373 MVA rating. Project DFAX: 13.8%.

### **3.2 Multiple Facility Contingency Analysis**

No additional overloads were identified for multiple facility contingencies.

### **3.3 Short Circuit and Stability**

Short circuit and stability analyses identified no additional network upgrades.

## **4. Required Network Upgrades and Cost Estimates**

| Facility                                             | Required Upgrade                            | Cost Estimate | Allocation to Project |
|------------------------------------------------------|---------------------------------------------|---------------|-----------------------|
| Limerick - East Weymouth 138kV                       | Rebuild line with higher-capacity conductor | \$4,863,000   | 61% (\$2,970,000)     |
| East Weymouth - Rec Shaw 138kV line                  | Rebuild line with higher-capacity conductor | \$2,647,000   | 61% (\$1,616,000)     |
| Lochbuie - Rec Shaw 138kV Line                       | Rebuild line with higher-capacity conductor | \$2,770,000   | 61% (\$1,691,000)     |
| Dowe Flats Substation - Curecanti 138kV line         | Rebuild line with higher-capacity conductor | \$2,812,000   | 82% (\$2,316,000)     |
| Alliant Sbd 9106 Rockwell Cr - Dowe Flats 138kV Line | Rebuild line with higher-capacity conductor | \$1,857,000   | 82% (\$1,530,000)     |
| Marin Creek - Alliant Sbd 9106 Rockwell Cr 138kV     | Rebuild line with higher-capacity conductor | \$11,005,000  | 82% (\$9,065,000)     |
| Clodine - Martin Creek 138kV                         | Rebuild line with higher-capacity conductor | \$3,211,000   | 82% (\$2,645,000)     |
| Eliza Crk 345/138kV XFMR                             | Replace transformer with higher-rated unit  | \$14,574,000  | 21% (\$2,995,000)     |
| Owatonna - Moser Generating Station 345kV            | Rebuild line with higher-capacity conductor | \$12,284,000  | 4% (\$488,000)        |
| Massena - Limerick 138kV                             | Reconductor line                            | \$1,701,000   | 61% (\$1,042,000)     |
| Marymount Substation - Massena 138kV L               | Reconductor line                            | \$1,468,000   | 61% (\$899,000)       |
| Marymount - Endeavor 138kV                           | Reconductor line                            | \$886,000     | 61% (\$543,000)       |
| Curecanti - Endeavor 138kV                           | Reconductor line                            | \$1,979,000   | 61% (\$1,212,000)     |
| Sprecher - Neset 138kV                               | Reconductor line                            | \$1,052,000   | 40% (\$422,000)       |
| Lochbuie - Moser Generating Sta 345kV Line           | Reconductor line                            | \$13,272,000  | 4% (\$514,000)        |
|                                                      | Total allocated to project                  |               | \$29,948,000          |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.